Game Design Document of Merchant Market

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Design 223

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Game Overview

Merchant Market is a game that will be developed around Sustainable Development Goal number 8, ‘decent work and economic growth’. It’s a game where players aim to create a sustainable business as a timber merchant. Players do this by trying to gain the most ‘fund’ and ‘planet’ tokens, this will also teach them how their actions can affect others in a market selling shared resources. The main age groups this will be targeted towards are high schoolers 13-18 years old.

The approximate playing time will be 30+ mins, depending on the amount of players, as this game can host 2-5 players. The difficulty of this game is 1.5/5 as it is relatively simple, but does require some thinking and strategising, as people can sabotage, and there are some random events that may become an obstacle.

The difficulty level is not too high so other age groups may be able to participate as well if they wish, due to the relative simplicity of this game. The difficulty level will also keep people engaged in the gameplay. It also has moving components and mechanics that involve other players, so players are almost always partially involved.

The objectives of this game are also clearly stated, and the starting player of the round rotates to try and give players fair opportunities. This game is designed to be more slow paced with some repetition so that players learn and absorb what the game is trying to teach them.

Background

The game board of most board games is typically made out of some type of solid card or cardboard. This is likely to be due to the flexibility of the material which makes it easy to fold and store away, and because it is more affordable in

comparison to other materials such as wood, especially when mass producing and commercialising. This is why we will also potentially create the main board out of similar materials.

The individual squares on the board of **Merchant Market** are roughly 4cm in width, thus making the game board relatively large in comparison to other games. This game will also use two standard D6 dice, because it is universal and can easily be replaced if lost.

The instruction manual will be made from paper, roughly A5 when folded, (originally A4). The paper will roughly be 80 GSM with a glossed finish, as most games typically have glossed instruction manuals. The character pieces and stands will be made out of cardboard for cost efficiency. The playing cards will be made out of card material, estimated GSM of 250, and have a matte finish with standard dimensions.

The game packaging will also likely be made of dense cardboard for durability. The dividers inside will potentially be made out of card with a relatively high GSM. The card sleeves will also be made out of card, estimated to be around 300 GSM.

Merchant Market's first prototypes will probably be made out of standard printing paper for cost and time efficiency, before exploring other material use for specific components.

Gameplay and Mechanics

Gameplay instructions

This game can be played by 2-5 players. The setting of this game is that players are timber merchants who have to share the same resource, timber. Each player takes turns rolling the dice and moving their character meeple around the board in a

clockwise direction before playing an action. Players must do one action unless otherwise stated by an event that was encountered on the board or through the event cards.

At the end of each round trees that the players had obtained will be exchanged for fund tokens. Players must also pay tax; the amount of tax needed to be paid depends on the round number. After each round the starting player of the round will shift to the person to the left of the previous person, see diagram below. The game ends after 10 rounds and the winner is whoever has the most “fund” and “planet” tokens combined. The game play will be written in the instruction manual in a step-by-step format.

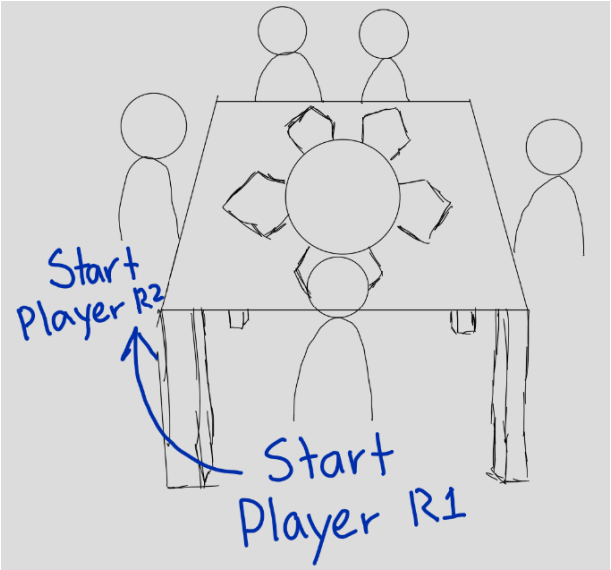


Diagram 1: Rotation of the starting player of each round. Own work

Game Components

- Two dice
- Twenty tree cards
- Plastic ziplock bag for token storage
- Fund tokens (41)

    - Sixteen $5 value fund tokens
    - Ten $3 value fund tokens
    - Fifteen $1 value fund tokens
- Planet tokens (41)
    - Sixteen planet tokens with value of 5
    - Ten planet tokens with value of 3
    - Fifteen planet tokens with value of 1
- Game box
- Game box dividers
- Card sleeves
- Character meeple (five)
- Game board
- Rules
- Event cards
    - Thirteen Personal
    - Seven Community
- Action pentagon lists (five)

Start set up

- Each player can get 10 fund tokens and 10 planet tokens.
- After rolling the dice, each player moves in a clockwise direction.
- There are tree cards as shared resources in each round in the centre of the board.
    - The amount of trees in the centre will vary depending on the number of players to simulate limited resources, see table below.

- (If the tree cards in the centre of the board run out in this round, the player who made the tree deck hit zero will lose 4 planet tokens)

Number of Players	Tree cards in the centre
2	5
3	8
4	11
5	14

Table 1: Tree card to player ratio

Tax end of each round

Round 1 - Tax $0

Round 2 - Tax $1

Round 3 - Tax $2

Round 4-10 - Tax $3

Actions each round

- Profitable act. -2 planet, +3 tree → exchanged into +$3
- Sustainable act. +1 planet, +1 tree → exchanged into +$1
- Replant act. +2 planet, -1 tree → pay $1, and put back a tree the next round you get tree/s (-$1 from tree exchange will be lost), the planet points gain is instant.

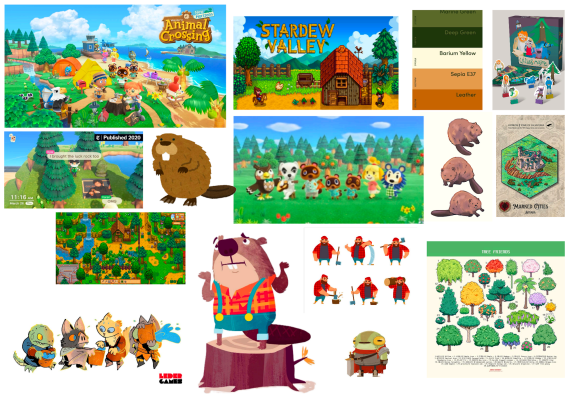
Mechanics

- You will be out of the game when your fund tokens reach zero.

- If there are no planet tokens, double the number of planets will be deducted from the profit.
- Roll die (D6), move around the game board.
- Each player can act once during a turn.
- Each tree turns into $1 at the end of each round.
- The player who gets the most funds + planet tokens will be the winner.

Game Concept Art

Mood board

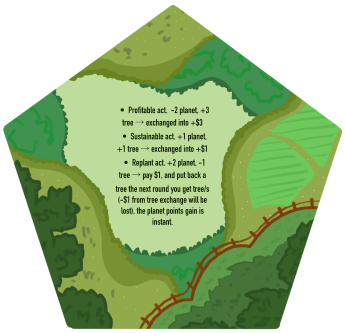


MOOCHI JELLY
DIACLONE
DIN ALTERNATE
CHALKBOARD SE

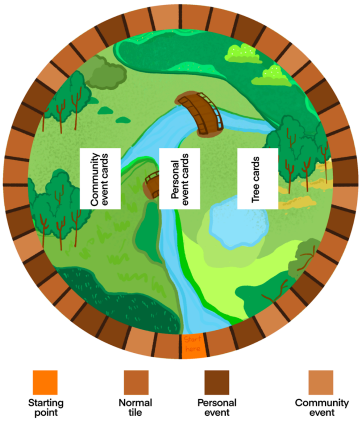
Images used listed under references

Game board

- Circle shape with a 50cm diameter
- Place for cards to go in the centre
- 40 squares - roughly 4 cm
- Squares are arranged around the edge of the decagon board, refer to the diagram, *Main game board* (next page).
- 20 event-triggering squares
- 5 detached pentagons with the action choices on them



Action pentagon. Own work



Main game board. Own work



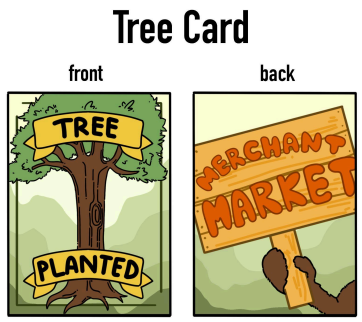
Game Box. Own work



Game pieces overview. Own work

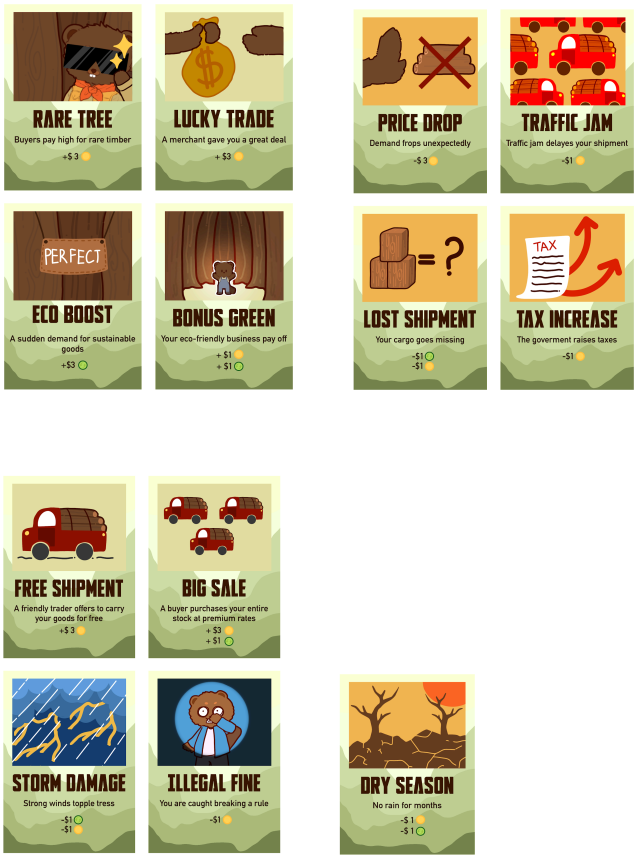
Cards (6.4 cm x 8.9 cm)

- 20 Tree cards



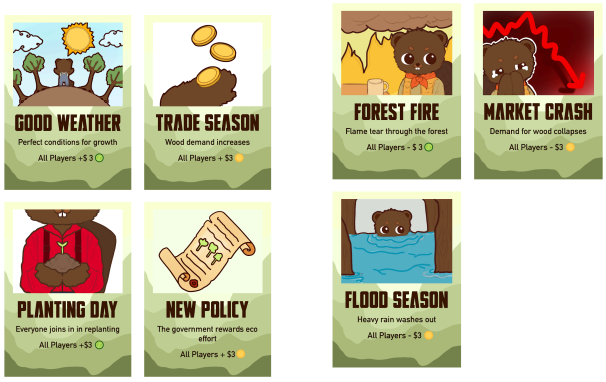
Tree card design. Own work

- 13 Personal event cards



Personal event cards. Own work

- 7 Community event cards



Community event cards. Own work

Tokens (diameter of 3cm)

- Fund tokens
    - Sixteen $5 value fund tokens
    - Ten $3 value fund tokens
    - Fifteen $1 value fund tokens
- Planet tokens
    - Sixteen planet tokens with value of 5
    - Ten planet tokens with value of 3
    - Fifteen planet tokens with value of 1



Fund & Planet tokens design. Own work

Character meeple

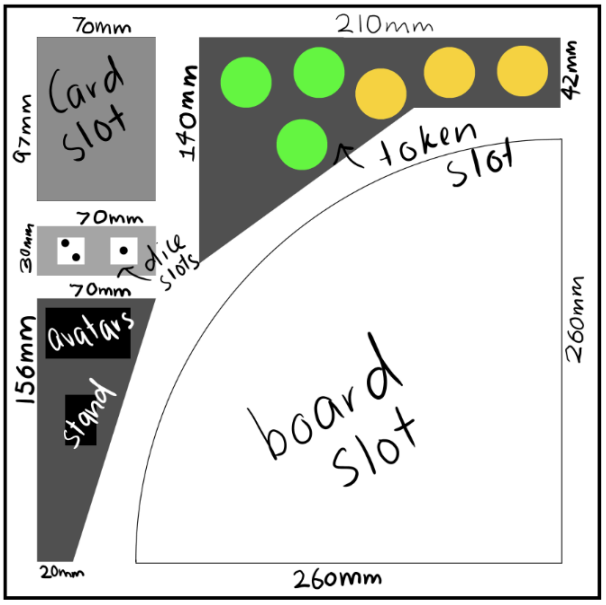
- Character stand (5)
- Character avatar (5cm x 3cm) (5 pieces)



Character token design. Own work

Packaging (refer to diagram)

- Game box (35cm x 35cm x 8cm)
- Dividers/indents
- Card sleeves (x3)
- Small plastic ziplock bag for token storage



Project Plan

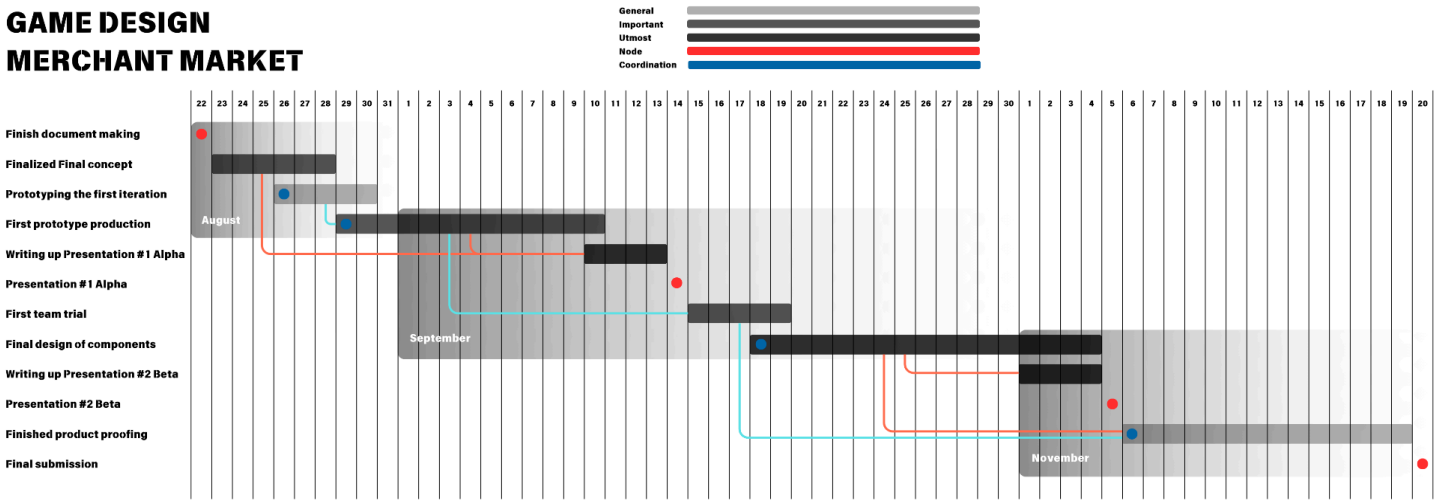
Inner packaging layout. Own work

Project Timeline

Our game will be developed in a 59-day cycle. The following is our project time design.

- 22/8 Finish document making
- 23/8 - 28/8 Finalised Final concept
    - Brainstorm, determine and adjust the final core game themes and mechanics
- 26/8 - 30/8 Planning the first prototype iteration of the game
    - Use simple materials such as paper/cards/notes to create low-fidelity prototypes, conduct rule trials, and improve the rules
- 29/8 - 10/9 Prototype production
    - Use the tools in Design Lab and cooperate with the skills of team members to create the first version of the game model, and record some improvements based on actual experience

- 10/9 - 13/9 Writing up Presentation #1 Alpha
- 14/9 Presentation #1 Alpha
- 15/9 - 19/9 The team plays the first version and collects feedback
    - Play the first prototype and get new balance and material feedback
- 18/9 - 4/10 Final design of art and components
    - Complete the final card illustrations, game board/component designs, and determine the final material and size
- 1/10 - 4/10 Write up Presentation #2 Beta
- 5/10 Presentation #2 Beta
- 6/10 - 19/10 Final proofing, final optimisation, and presentation.
    - Print a high-fidelity version for small-scale game testing. Follow the modification instructions to complete the final version and prepare for presentation.
- 20/10 Final submission



Team division of labor

Planning Director: Gemma Wong

- Plan the project progress and delegate team members tasks

Mechanism Designer: Gemma Wong / Yicheng Mao

- Responsible for the game mechanism design and fairness adjustment

Art Design: Ellie Nguyen / Nicky Ning / Yicheng Mao

- Design the cards and bases needed for the game

Component production: Xinyang Ji **/** Ellie Nguyen (Gemma Wong extra help)

- Responsible for producing all the parts needed for the physical game

Testing and Feedback: Xinyang Ji / Nicky Ning

- Organise trial activities, record player feedback, and assist in rule iteration

Possible difficulties and solutions

- The rules are not balanced enough: Collect user feedback through multiple in-group/small-scale public trials and improve the game mechanism based on the feedback
- Art is out of touch with the theme: Ensure that icons and text clearly and intuitively convey the theme of the game, adopt a unified visual standard, and pay more attention to art production
- Difficulty in material proofing: In the early stage, easy-to-obtain simple materials are used to proof-proof the game, focusing on the mechanism. In the later stage, the content is planned in detail before closing. Each component is proofed in turn to facilitate modification.
- Too little team communication: Contact team members more often through communication software. Questioning and commenting on the production of the game is also part of progress.

Production workflow

We use an iterative development process, optimising from paper to final product:

- Pre-production planning → Idea, theme, target audience, core mechanics
- Prototyping → Test gameplay with simple materials
- Testing and iteration → Playtests, rule fixes
- Art and layout → Card illustrations, board design, instruction manual
- Product proofing → Printing a near-final version
- Final revisions → Small-scale playtests, final revisions, and presentations

Pre-production planning → Prototyping → Testing and iteration → Art and layout → Product proofing → Final revisions

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